

(Please write your Exam Roll No. immediately)

Exam Roll No: _____

END TERM EXAMINATION
THIRD SEMESTER [B.TECH.] FEBRUARY, 2023
Paper Code: CIC-205 • Subject: Discrete Mathematics

Time: 3 Hours

Maximum Marks: 75

Note: Attempt **five** questions in all including **Q. No. 1** which is compulsory. Select **one** question from each unit. Assume missing data, if any.

Q. No.	Question	Max. Marks	CO(s)
Q1 (a)	Represent the statement using predicate and quantifier and negate it: "For all real numbers x , if $x > 5$ then $x^2 > 25$."	[3]	CO1
Q1 (b)	Evaluate the condition of a function to be surjective (onto). Give an illustrative example.	[3]	CO2
Q1 (c)	Find the converse and contrapositive of the statement: "If x is positive then $x \neq 0$."	[3]	CO1
Q1 (d)	Define an Euler path and an Euler circuit with the help of an example graph.	[3]	CO4
Q1 (e)	Prove that the set of all integers $\mathbf{Z}$ does not form a group under multiplication with identity element 1.	[3]	CO3
UNIT – I			
Q2 (a)	Draw the following set operations with the help of Venn diagrams: (i) Union ($A \cup B$) (ii) Intersection ($A \cap B$) (iii) Disjoint Sets ($A \cap B = \emptyset$) (iv) Set Difference ($A \setminus B$)	[4]	CO1
Q2 (b)	Prove that if xy is odd, then both x and y are odd, using indirect proof by contrapositive.	[4]	CO1
Q2 (c)	Show that $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ using rules of logical equivalence and propositional calculus.	[7]	CO1
OR			
Q3 (a)	Prove that if x^2 is even, then x is even (where $x \in \mathbf{Z}$).	[4]	CO1
Q3 (b)	Check the validity of the following argument using rules of inference: "Either the races are fixed or the casinos are crooked. If the casinos are crooked, then the tourist trade will decrease. If the tourist trade decreases, then the police will be happy. The police force is not happy. Therefore, the races are fixed."	[4]	CO1
Q3 (c)	In how many ways can a team of 11 cricketers be chosen from 6 bowlers, 4 wicket-keepers and 11 batsmen to give a majority of batsmen if at least 4 bowlers are to be included and there is exactly one wicket-keeper?	[7]	CO1
UNIT – II			
Q4 (a)	Analyse all the three cases of Master's method for solving divide-and-conquer recurrence relations of the form $T(n) = aT(n/b) + f(n)$.	[7]	CO2
Q4 (b)	Minimize the given Boolean function using a 4-variable K-map: $F(A, B, C, D) = ABC'D' + ABC'D + AB'C'D + ABCD + AB'CD + ABCD' + AB'CD'$ Note: Prime notation (') denotes complementation. Equivalent to minterms $\Sigma m(9, 10, 11, 12, 13, 14, 15)$.	[8]	CO2
OR			
Q5 (a)	Let $A = \{1, 2, 3, 4, 6\}$ and R is a relation on set A such that aRb if $a \mid b$ (a divides b). Check whether relation R is a POSET and draw its Hasse diagram.	[7]	CO2
Q5 (b)	What are the different solution methods for first order recurrence relations with constant coefficients? Explain homogeneous and particular solutions with an illustrative example.	[8]	CO2

Q. No.	Question	Max. Marks	CO(s)
UNIT – III			
Q6 (a)	Show that if $a^2 = e$ for all elements a in a group $(G, *)$, then G is an abelian (commutative) group.	[5]	CO3
Q6 (b)	Explain Group Homomorphism, Isomorphism, and Automorphism with clear mathematical definitions and examples.	[5]	CO3
Q6 (c)	State and prove Lagrange's Theorem for finite groups.	[5]	CO3
OR			
Q7 (a)	Analyse the necessary and sufficient conditions for a group to be abelian. Provide one example of a non-abelian group.	[5]	CO3
Q7 (b)	State Cayley's Theorem and explain its fundamental importance in group theory representation.	[5]	CO3
Q7 (c)	If for each a and b in a group G , $(ab)^2 = a^2b^2$, show that G must be an abelian group.	[5]	CO3
UNIT – IV			
Q8 (a)	Evaluate the total number of proper vertex colorings of a Complete Graph K_n using k colors such that no two adjacent vertices have the same color. Derive the chromatic polynomial.	[7]	CO4
Q8 (b)	State and prove the Breadth First Search (BFS) algorithm. Explain how it finds the shortest path in an unweighted graph with an illustrative example.	[8]	CO4
OR			
Q9 (a)	Write a short note on Minimum Spanning Tree (MST): Describe Kruskal's algorithm and Prim's algorithm, detailing their step-by-step execution and time complexities.	[8]	CO4
Q9 (b)	Write a short note on the Five Color Theorem: State the theorem for planar graphs and explain the inductive proof method using Kempe chains.	[7]	CO4